Common Blocking and Span Equality in the Non-periodic MPS Fundamental Theorem

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1 The compressed source argument

The proof of the non-periodic matrix product state fundamental theorem passes from arbitrary tensors with the same matrix product vectors to basis-of-normal-tensors comparison data. The source argument is the MPDO canonical-form paper [CPGSV16, CPGSV17], together with the later review [CPGSV21].[1] The cited papers are not being questioned. The point is that a formal proof has to separate several canonical-form steps which are compressed in the source prose.

For a tensor $A = (A^i)_{i=1}^d$, the source paper defines the matrix product vectors by

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=1}^{d} \mathrm{tr}(A^{i_1}\cdots A^{i_N})|i_1\rangle \otimes \cdots \otimes |i_N\rangle,$$

for $N = 1, 2, 3, \dots$ [CPGSV16, Definition `MPV`, lines 130–138]. After removing invariant subspaces, the same paper writes the tensor in block form

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i,$$

allowing zero blocks and scaling the nonzero blocks so that their transfer maps have spectral radius one. Each nonzero irreducible block may still have peripheral period. Blocking by a common multiple of the periods removes that periodicity [CPGSV16, lines 214–231]. The review follows the same canonical-form argument [CPGSV21, lines 1798–1820].

The later comparison is made through bases of normal tensors. A basis of normal tensors is a minimal family $(A_j)_{j=1}^g$ of normal tensors such that every matrix product vector of A is a linear combination of the matrix product vectors of the A_j, and such that those vectors are eventually linearly independent. Equivalently, every normal block in the canonical form is a phase times a gauge transform of exactly one basis tensor, with the family chosen

[1] For traceability, this note corresponds to issue #1047.

minimally [CPGSV16, Proposition `prop:char-BNT`, lines 271–280]. The review states the same characterization [CPGSV21, Proposition `prop:char-BNT`, lines 1846–1859]. For a tensor in canonical form one then has an expansion

$$|V^{(N)}(A)\rangle = \sum_{j=1}^{g} \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) |V^{(N)}(A_j)\rangle.$$

This is the basis-of-normal-tensors expansion used in the equal-case proof [CPGSV16, lines 283–302]. The review uses the same expansion [CPGSV21, lines 1864–1885].

Injectivity enters at a different point. The source first defines injective and block-injective canonical form, and then invokes the quantum Wielandt theorem to say that, after a bounded further blocking, every normal tensor becomes injective [CPGSV16, lines 317–332]. The review records the corresponding $3D^5$ bound for block-injective canonical form [CPGSV21, lines 2120–2124]. This step is not the same as period removal: period removal gives primitive normal blocks, while the overlap-rigidity arguments used later require one-site injectivity, or an explicit replacement after further blocking.

Finally, the source comparison for the equal case is compact. Equality or proportionality of the matrix product vector families first matches the two bases of normal tensors by phase and gauge. After relabeling, the coefficient multiplicities are compared through the power sums

$$\sum_{q=1}^{r_{a,j}} \mu_{j,q}^N = \sum_{q=1}^{r_{b,j}} (\nu_{j,q} e^{i\phi_j})^N, \qquad \text{for all } j \text{ and } N.$$

The elementary power-sum lemma then recovers the multiplicities and the weights, and the blockwise gauges are assembled into the global conjugating matrix [CPGSV16, lines 1181–1192]. The review states the resulting proportional and equal MPV fundamental theorems in the same form [CPGSV21, Theorem `thm1` and Corollary `II_cor2`, lines 1891–1900].

2 The formal argument

The formal equality relation $\mathsf{SameMPV}_2$ compares matrix product vector coefficients for all finite words, including the empty word. This convention is stronger than the positive-length family in the source definition. An all-zero block of bond dimension D_0 contributes nothing at positive length, but it contributes D_0 at length zero, because the empty product has trace D_0. Consequently the formal proof separates the zero-tail contribution from the nonzero part, proves positive-length equality for the nonzero parts, and records the exact $N = 0$ identity separately.[2]

[2]See `blueprint/src/chapter/ch08_canonical.tex:967--1022` and `blueprint/src/chapter/ch08_canonical.tex:`

The common blocking step is also more explicit. For a single family of blocks, the source phrase "block by the least common multiple of the periods" hides no mathematical difficulty. In the two-tensor comparison, however, the formal proof must choose one positive physical blocking length for both sides. For each original block with period-removal length m_k, the common length is written as $p = m_k e_k$. The tensor obtained by first blocking by m_k and then by e_k has physical labels given by nested words, while the tensor obtained by blocking once by p has physical labels in the alphabet of words of length p. The formal statement therefore keeps the relabeling between these two alphabets as part of the statement.[3] This is why the canonical-form theorem supplies relabeled common-sector families and a conditional label-compatibility statement, rather than immediately replacing every canonical nonzero part after blocking by its flattened sector tensor.

The formal comparison theorems downstream use primitive overlap-rigidity data for sector bases. Those data include equality of the finite-length spans of the basis matrix product vectors and one-site injectivity of the basis blocks. Thus the additional Wielandt blocking needed for injectivity remains a separate refinement after period removal. Treating it separately prevents a common period-removal length from being used as though it already supplied the injective hypotheses needed by the overlap-span theorem.[4]

The paper's compact BNT matching and power-sum comparison is represented in the formal statement by common phase-cover and finite-length span-equality adapters. A common phase cover consists of a common family of nonzero blocks, surjective maps from the two block families onto that common family, and phase equivalences between each block and its common representative. Such data imply that, for every system size, the spans of the two nonzero-block MPV families are equal. When a BNT block-permutation gauge-phase comparison has already produced a permutation, matched bond dimensions, and gauge phases, the formal construction gives the common phase cover from that comparison and then obtains the span equality needed by the sector comparison theorem.[5] These lemmas state precisely what comparison data are

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`3036--3164`, with labels `def:zero_mps_tensor`, `thm:mpv_zero_tensor`, `thm:zero_block_separation`, `thm:nonzero_block_block_power_zero_tail_identity`, and `thm:nonzero_block_same_mpv_zero_tail_eq`. The corresponding zero-block cancellation step in the final comparison chapter is `blueprint/src/chapter/ch11_fundamental_theorem_proof.tex:1202--1217`.

[3] The common-sector data and the relabeling results are recorded in `blueprint/src/chapter/ch08_canonical.tex:1645--1824` and `blueprint/src/chapter/ch08_canonical.tex:3230--3284`. The main formal statement is `MPSTensor.fundamentalTheorem_after_blocking_commonLength_commonSector`, with the one-sided relabeling-compatible zero-tail statement `MPSTensor.zeroTail_commonFlat_of_reindexed`; see `TNLean/MPS/CanonicalForm/SectorComparison/CommonSectorData.lean` and `TNLean/MPS/CanonicalForm/SectorComparison/CommonSectorTransport.lean`.

[4] The span hypotheses are stated in `blueprint/src/chapter/ch11_fundamental_theorem_proof.tex:590--602`, label `def:sector_basis_overlap_span_hyps`. The remaining canonical proof obligations are summarized in `blueprint/src/chapter/ch08_canonical.tex:3289--3341`.

[5] See `blueprint/src/chapter/ch11_fundamental_theorem_proof.tex:75`

needed; they do not by themselves derive all of that data from an arbitrary equality of total MPV families.

3 Why these distinctions belong together

These distinctions are coupled in the single conversion from an arbitrary equality of MPV families to the hypotheses of the BNT sector-comparison theorem. The zero-tail convention determines what is known at length zero after each blocking. The common blocking construction determines the physical alphabet and the relabeled sector families to which the later comparison must apply. The injectivity refinement may further change the final block length before the overlap-rigidity hypotheses are available. The common phase-cover or BNT block-permutation gauge-phase data then have to be constructed for those final nonzero-sector families, not for an earlier family before relabeling or before injectivity blocking.

For this reason common blocking and finite-length span equality should be kept in one account. The formal proof needs the whole chain: remove the zero tail, choose one common physical blocking length, account for the word-relabeling, keep the later injectivity blocking separate, and then turn BNT matching data into the finite-length nonzero-sector span equality used by the sector comparison theorem. The blueprint states this chain as the remaining connection between the canonical-form chapter and the final comparison chapter.[6]

This note should also not be read as a claim that the same-side equal-norm scalar obstruction discussed in the broader fundamental-theorem work is a flaw in the CPSV or CPGSV arguments. That obstruction corrected an over-strong formal statement which tried to identify an equal-norm class with a single basis tensor. The BNT expansion in the papers already allows multiple basis tensors and multiplicities. The formal argument therefore moved from naive norm-class grouping to phase-class sector decompositions and common-cover data.

`5--923`. The corresponding formal objects include the common phase-cover structure and span lemmas, the block-span and common-phase-cover overlap-span adapters, and the block-permutation gauge-phase specialization. They are located at `TNLean/MPS/CanonicalForm/PhaseCover.lean` and `TNLean/MPS/CanonicalForm/PhaseClassSectorData.lean`; the main adapter names have prefix `MPSTensor.exists_bnt_sectorDecomp_pair_with_overlapSpan_of_` and suffixes `block_span_eq` and `commonPhaseCover`; the block-permutation gauge-phase conclusion first gives span equality through MPSTensor.mpv˙span˙eq˙of˙blockPermutationGaugePhaseConclusion.

[6] See the "Remaining hypotheses" discussion in `blueprint/src/chapter/ch11_fundamental_theorem_proof.tex:1277--1341`. Audit notes recording the same remaining obligations include `audits/2026-04-28_issue969_direct_common_blocking.md:139--153`, `audits/2026-04-28_issue970_common_phase_cover.md:81--95`, `audits/2026-04-29_issue942_common_exact_nonzero_sector.md:66--97`, and `audits/2026-04-29_issue944_overlap_span_from_common_live.md:83--94`.

4 Conclusion

The conclusion is that the cited papers compress several standard canonical-form operations; no counterexample is claimed. Zero or nilpotent contributions are invisible to positive-length MPVs, periods are removed by blocking, injectivity is obtained by a later Wielandt-type blocking, BNT bases are matched by phase and gauge, and coefficient multiplicities are recovered from power sums. The formal development exposes these operations as separate statements because the formal MPV relation includes the empty word, the two sides must be placed over one explicitly relabeled alphabet after blocking, and the downstream comparison theorem requires finite-length span equality and injective final blocks.

Thus the comparison gives neither a counterexample nor a theorem-level source error. It explains why the formal proof cannot compress the argument into a single invocation of the BNT uniqueness theorem. The relevant comparison points are zero-tail cancellation, common primitive block construction, common blocking, finite-length span equality, zero-tail transport, overlap-span hypotheses, and iterated-blocking relabeling.[7]

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

[CPGSV17] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. *Annals of Physics*, 378:100–149, 2017.

[CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.

[7] For traceability, these are recorded in issue #652, issue #942, issue #969, issue #970, issue #971, issue #944, and issue #990.